

PREDICTED CLINICAL DIAGNOSIS
ADENOCARCINOMA

DIAGNOSTIC CONFIDENCE
91.4%
DenseNet-121 Softmax

1. Imaging Evidence & XAI Localization Map



Figure 1A: Raw Input Chest CT Slice (256×256)

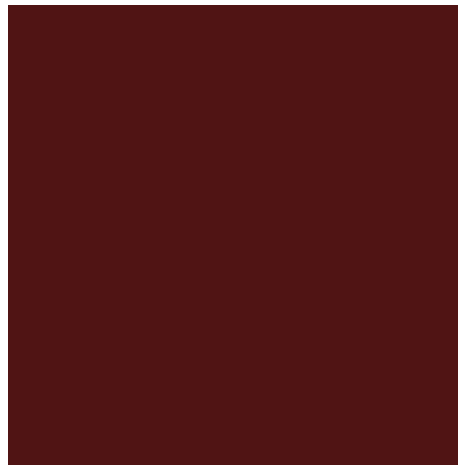


Figure 1B: Grad-CAM Activation Heatmap Overlay

2. Heatmap Feature Extraction (Quantitative Insights)

Metric / Parameter	Measurement / Value	Clinical Significance
Anatomical Location	Upper Right Lung	Primary lesion centroid in lung coordinate grid
Activation Level	High	Relative gradient magnitude on final dense layer
Region Size	18.2% of lung area	Proportion of segmented parenchyma involved
Activation Intensity	0.82 (High)	Peak normalized attribution score (0.0 – 1.0)
Spatial Distribution	High (Focal)	Dispersion cluster radius around lesion epicenter

3. Natural Language Explanation (NLP Module)

Primary Finding:

"The model predicts Adenocarcinoma with 91.4% confidence. The prediction was primarily influenced by a highly activated region in the upper right portion of the lung, which occupies about 18% of the lung area."

Formal Radiological Impression:

IMPRESSION: Findings consistent with Adenocarcinoma. Saliency mapping localizes the primary diagnostic determinant to the Upper Right Lung (occupying ~18.2% of segmented lung volume) with focal concentration and peak intensity 0.82.

Recommended Next Steps:

Correlate with contrast-enhanced chest CT / PET-CT targeting the Upper Right Lung lesion.

4. Multi-Class Probability Breakdown

Class Name	Probability	Confidence Bar
Adenocarcinoma	91.40%	
Squamous Cell Carcinoma	5.10%	
Large Cell Carcinoma	2.10%	
Normal	1.40%	

CONFIDENTIAL MEDICAL REPORT — BlackBoxMD-XAI is an artificial intelligence-assisted diagnostic research tool. All visual explanations and automated findings must be clinically verified by a licensed radiologist or oncologist before making diagnostic or therapeutic decisions.